

Spotify Descriptive Statistics

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STAT 3120

Statistical Analysis of Spotify Dataset.

Setup

```
#install.packages(c("readxl", "dplyr", "ggplot2", "janitor", "epitools", "gt", "car"))
```

```
library(readxl)
library(dplyr)
library(ggplot2)
library(janitor)
library(epitools)
library(gt)
library(car)
```

```
# Export Dataset, substitute "path/to/file" with actual path
sp <- read_excel("Data/spotify_stat3120.xls")
# View(sp) # To view dataset in R
```

Descriptive Statistics and Cleaning

Danceability (Quantitative)

```
dance <- sp$danceability
summary(dance) # min 0.166, Q1 0.511, med 0.636, mea 0.629, Q3 0.754, max 0.954
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.166	0.511	0.636	0.629	0.754	0.954

```
sd(dance)      # 0.159
```

```
[1] 0.1594543
```

Energy (Quantitative)

```
energy <- sp$energy
summary(energy) # min 0.109, Q1 0.643, med 0.823, mea 0.778, Q3 0.935, max 0.998
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1090	0.6430	0.8230	0.7782	0.9350	0.9980

```
sd(energy)      # 0.178
```

```
[1] 0.177742
```

Loudness (Quantitative, cleaning required)

```
clean_sp <- filter(sp, loudness < 0) # remove values greater than 0
loud <- clean_sp$loudness
var(loud)      # 8.166
```

```
[1] 8.165856
```

```
sd(loud)      # 2.858
```

```
[1] 2.857596
```

```
summary(loud) # min -17.912, Q1 -8.063, med -6.126, mea -6.471, Q3 -4.737, max -0.344
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-17.912	-8.063	-6.126	-6.471	-4.737	-0.344

```
remaining_loud <- (length(sp$loudness) - length(loud)) # 5 values were removed
remaining_loud
```

```
[1] 3
```

Tempo (Quantitative)

```
tempo <- sp$tempo
summary(tempo) # min 81.99, Q1 129.98, med 145.00, mea 147.22, Q3 160.18, max 220.04
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
81.99	129.98	145.00	147.22	160.18	220.04

```
sd(tempo) # 23.244
```

```
[1] 23.24385
```

Qualitative Variables

```
table(sp$genre)
```

Dark Trap	dnb	Emo	hardstyle	Hiphop	Pop	psytrance	Rap
43	38	20	48	38	2	50	17
RnB	techhouse	techno	trance	trap	Trap	Meta	Undergrou
20	44	32	40	38	34	79	

```
# Dark Trap 43, DnB 38, Emo 20, Hardstyle 48, HipHop 38, Pop 2, Psytrance 50, Rap 17
# RnB 20, Techhouse 44, Techno 32, Trance 40, Trap 38, Trap M 34, Underground 79.
```

Creating DanceCat by separating Danceability by the mean

```
sp$dancecat <- ifelse(sp$danceability > .629, "Danceable", "Not Danceable")
```

Removing Pop because of low count of 2

```
clean_sp <- sp %>% filter(genre != "Pop")
```

Contingency table and expected values

```
# New contingency table
tab <- table(clean_sp$dancecat, clean_sp$genre)
tab
```

	Dark Trap	dnb	Emo	hardstyle	Hiphop	psytrance	Rap	RnB	techhouse
Danceable	21	8	1	3	20	22	15	7	41
Not Danceable	22	30	19	45	18	28	2	13	3

	techno	trance	trap	Trap	Meta	Undergrou
Danceable	27	9	14	20		69
Not Danceable	5	31	24	14		10

```
# Expected values
expected <- chisq.test(tab)$expected
expected
```

	Dark Trap	dnb	Emo	hardstyle	Hiphop	psytrance
Danceable	22.01664	19.45656	10.240296	24.57671	19.45656	25.60074
Not Danceable	20.98336	18.54344	9.759704	23.42329	18.54344	24.39926

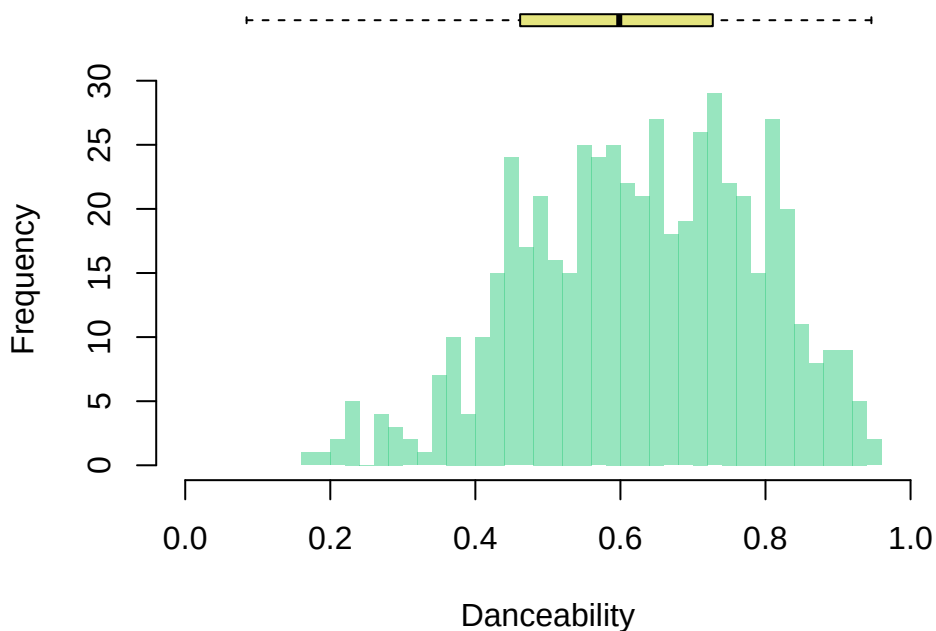
	Rap	RnB	techhouse	techno	trance	trap
Danceable	8.704251	10.240296	22.52865	16.38447	20.48059	19.45656
Not Danceable	8.295749	9.759704	21.47135	15.61553	19.51941	18.54344

	Trap	Meta	Undergrou
Danceable	17.4085	40.44917	
Not Danceable	16.5915	38.55083	

Graphs

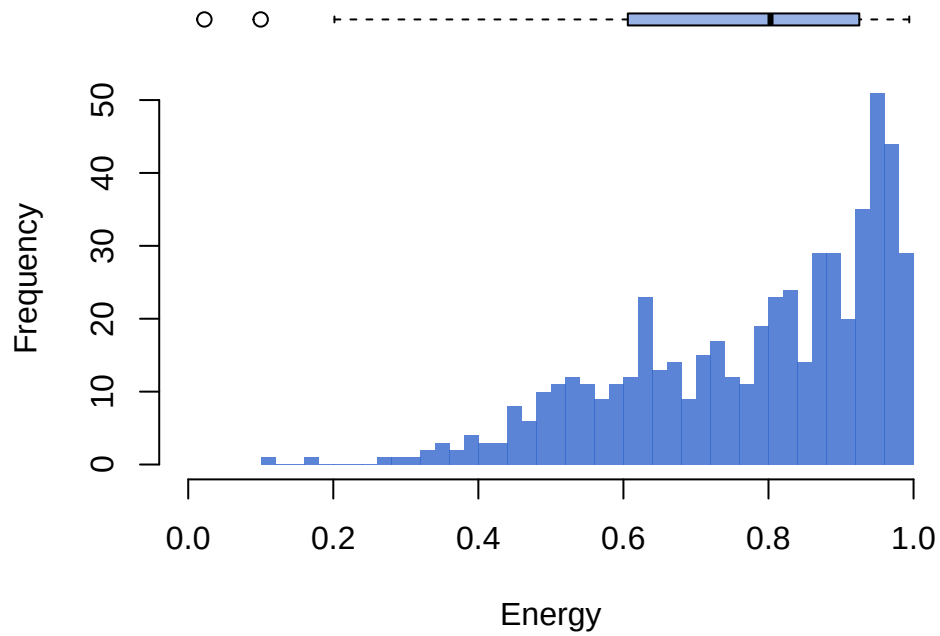
Boxplot with histogram for Danceability

```
layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(dance, horizontal=TRUE, ylim=c(0,1), xaxt="n",
        col=rgb(0.8,0.8,0,0.5), frame=F)
par(mar=c(4, 5, 1.1, 2.1))
hist(dance, breaks=40, col=rgb(0.2,0.8,0.5,0.5), border=F,
     main="", xlab="Danceability", ylab="Frequency", xlim=c(0,1))
```



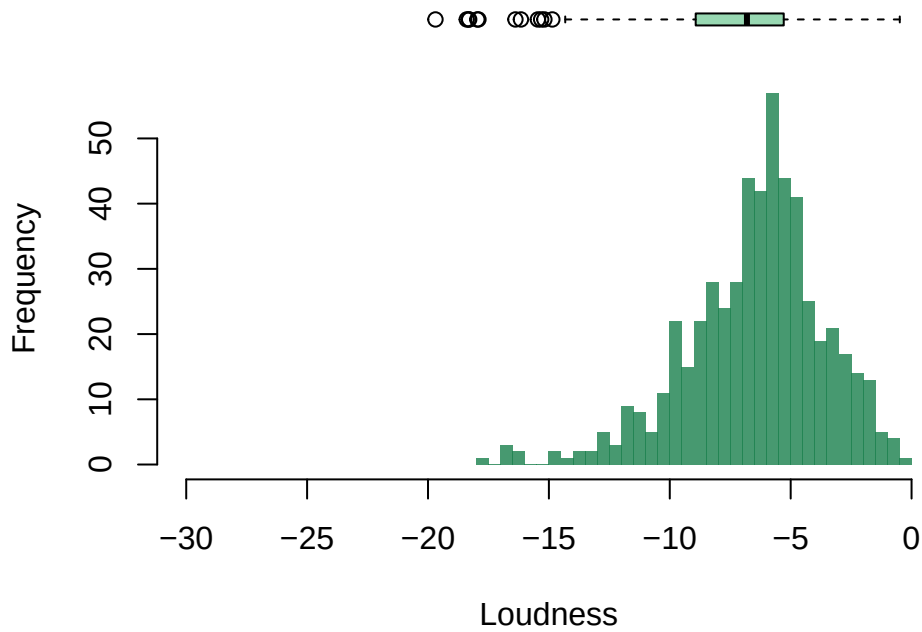
Boxplot with histogram for Energy

```
layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(energy, horizontal=TRUE, ylim=c(0,1), xaxt="n",
        col=rgb(0.2, 0.4, 0.8, 0.5), frame=F)
par(mar=c(4, 5, 1.1, 2.1))
hist(energy, breaks=40, col=rgb(0.2, 0.4, 0.8, 0.8), border=F,
     main="", xlab="Energy", ylab="Frequency", xlim=c(0,1))
```



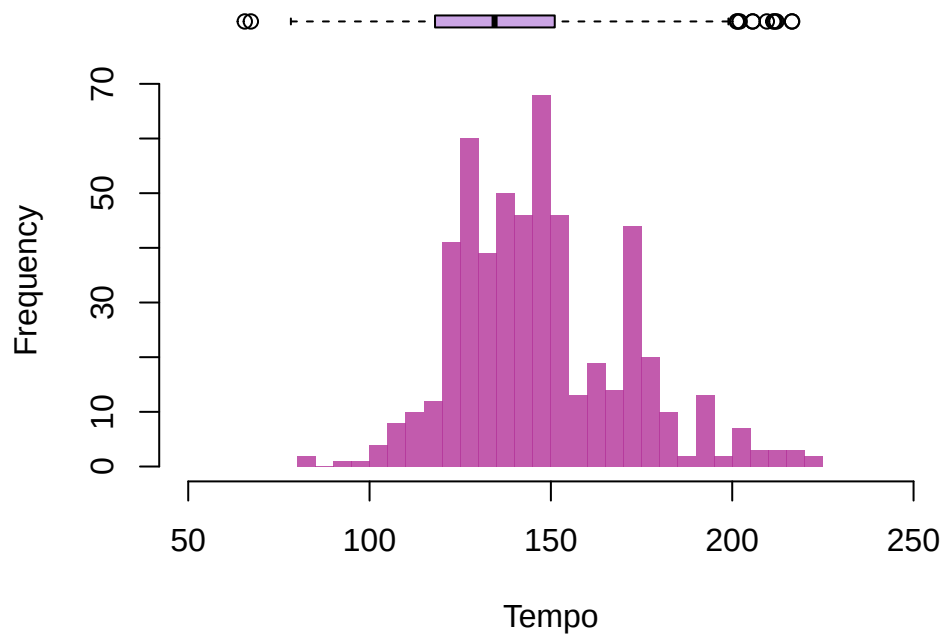
Boxplot with histogram for Loudness

```
layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(loud, horizontal=TRUE, ylim=c(-30,0), xaxt="n",
        col=rgb(0.2, 0.7, 0.4, 0.5), frame=F)
par(mar=c(4, 5, 1.1, 2.1))
hist(loud, breaks=40, col=rgb(0.1, 0.5, 0.3, 0.8), border=F,
     main="", xlab="Loudness", ylab="Frequency", xlim=c(-30,0))
```



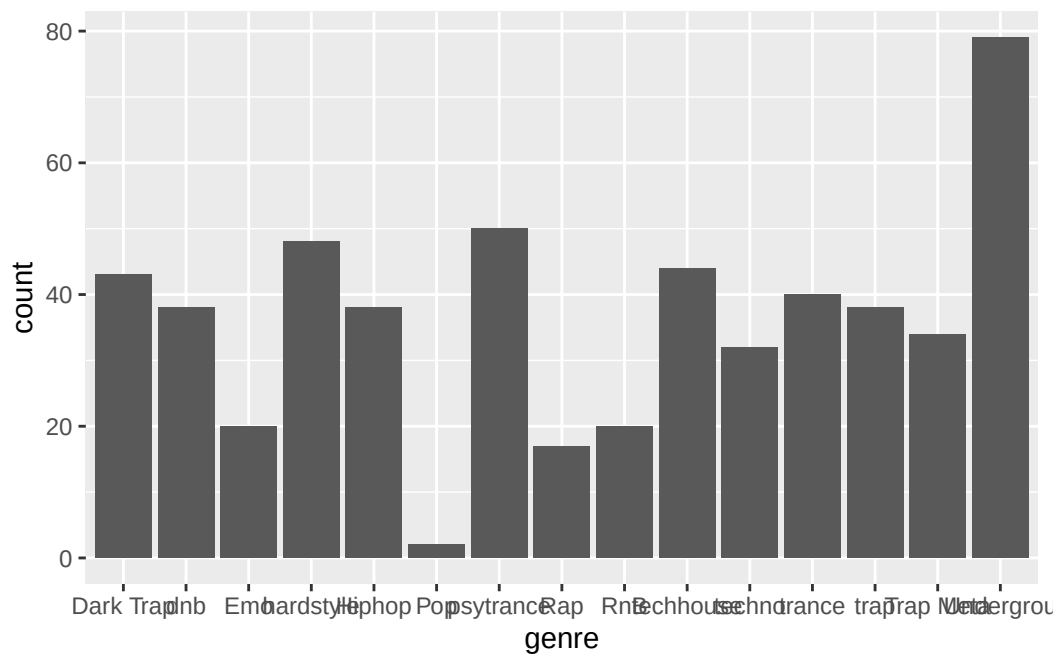
Boxplot with histogram for Tempo

```
layout(mat = matrix(c(1,2),2,1, byrow=TRUE), height = c(1,8))
par(mar=c(0, 3.1, 1.1, 2.1))
boxplot(tempo, horizontal=TRUE, ylim=c(50,250), xaxt="n",
        col=rgb(0.6, 0.3, 0.8, 0.5), frame=F)
par(mar=c(4, 5, 1.1, 2.1))
hist(tempo, breaks=40, col=rgb(0.7, 0.2, 0.6, 0.8), border=F,
     main="", xlab="Tempo", ylab="Frequency", xlim=c(50,250))
```

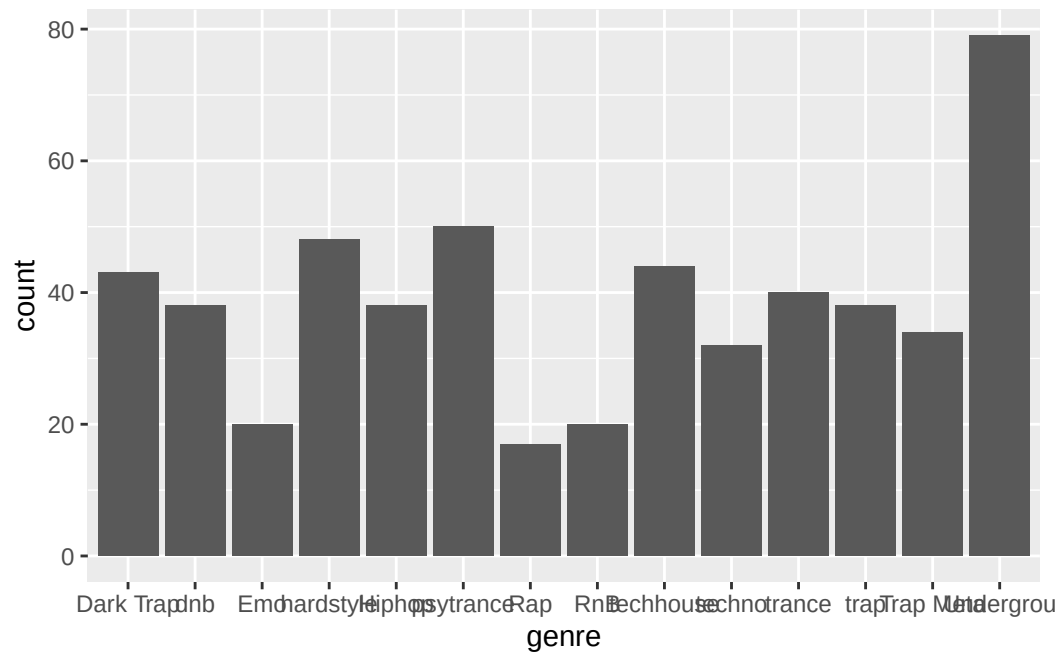


Bar plot for Genre

```
ggplot(sp, aes(x = genre)) + geom_bar()
```

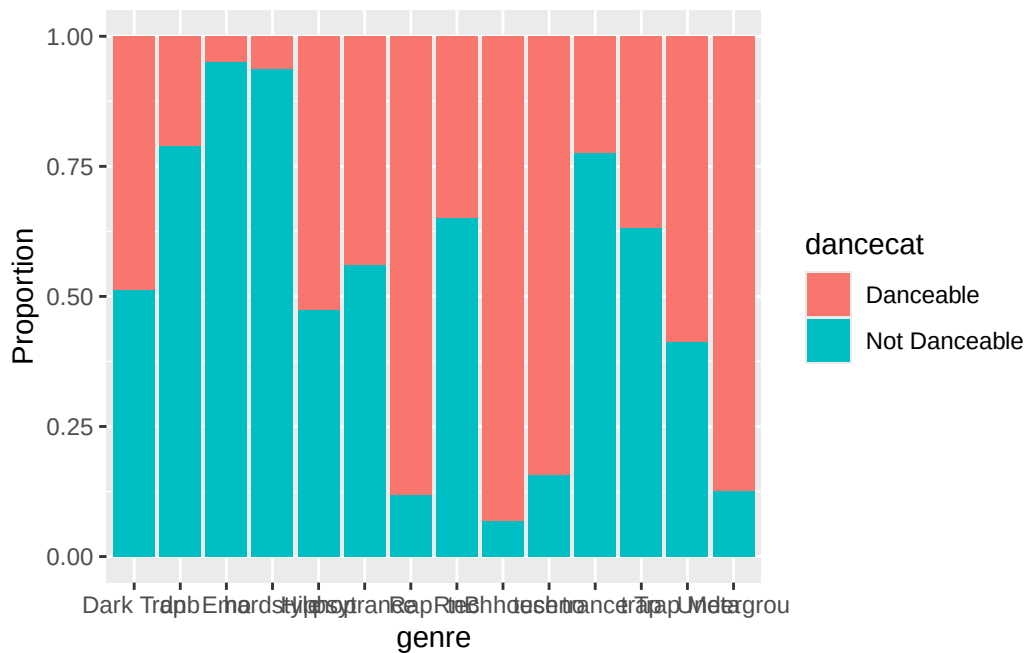



```
# without pop
ggplot(clean_sp, aes(x = genre)) + geom_bar()
```



Stacked Proportional bar chart (genre vs dance)

```
ggplot(clean_sp, aes(x = genre, fill = dancecat)) +
  geom_bar(position = "fill") +
  labs(y = "Proportion")
```



Statistical Tests

Test 1: Multiple Linear Regression Analysis

```
# Regression Test
model <- lm(danceability ~ loudness + tempo, data = sp)
summary(model) # df=540, f-stat=25.09, p=3.803e-11
```

Call:

```
lm(formula = danceability ~ loudness + tempo, data = sp)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.46147	-0.10904	0.00733	0.11471	0.38844

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.7584862	0.0474687	15.979	< 2e-16 ***
loudness	-0.0103119	0.0022990	-4.485	8.90e-06 ***
tempo	-0.0013299	0.0002865	-4.641	4.35e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1528 on 540 degrees of freedom

Multiple R-squared: 0.08503, Adjusted R-squared: 0.08164

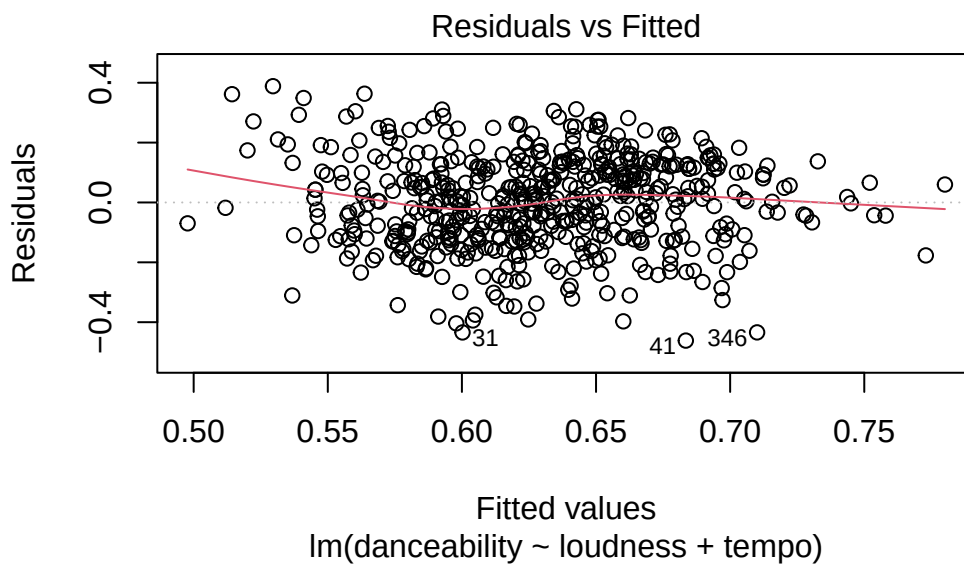
F-statistic: 25.09 on 2 and 540 DF, p-value: 3.803e-11

Checking conditions

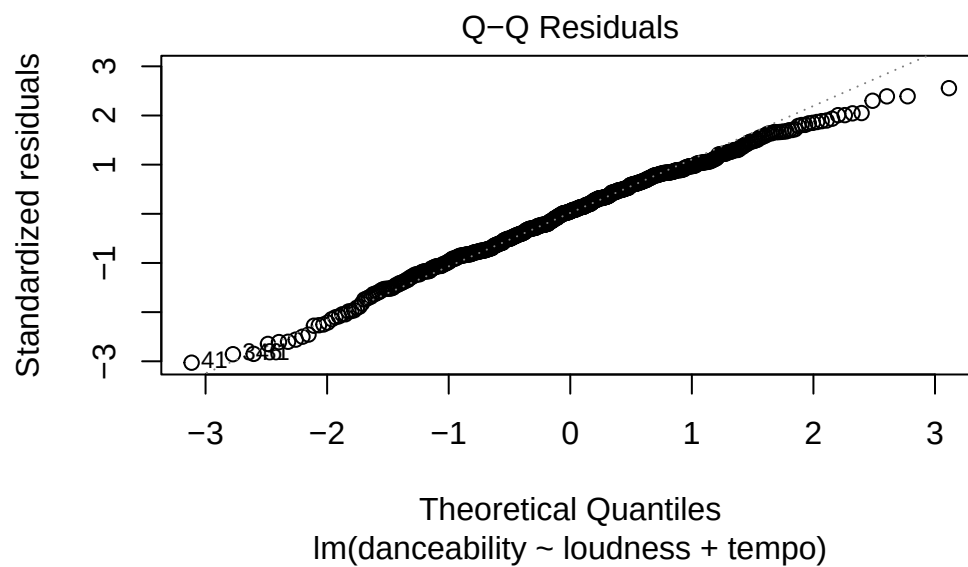
```
vif(model) # Multicollinearity check, passed bc close to 1.
```

```
loudness    tempo  
1.029716 1.029716
```

```
plot(model, which = 1) # Residual plot, passed graph looks decently centered around the middle
```



```
plot(model, which = 2) # Q-Q plot, passed no jarring signs.
```



Test 2: Chi-Square Test of Independence

```
# Verifying expected counts
all(expected >= 5) # true, satisfied
```

```
[1] TRUE
```

```
# Chi Square Test-of-Independence
chi_result <- chisq.test(tab)
chi_result # x-squared=186.87, df=13, p<2.2e-16
```

Pearson's Chi-squared test

```
data: tab
X-squared = 185.87, df = 13, p-value < 2.2e-16
```